Docker containers, I assume that you have Puppet master and agents configured properly.

Setting up the Puppet master

1. Docker installation on the Puppet master requires the *puppetlabs-stdlib* library module. Install the required Puppet module as shown below:

```
#puppet module install puppetlabs/stdlib
Notice: Preparing to install into /etc/puppet/modules...
Notice: Downloading from https://forgeapi.puppetlabs.com...
Notice: Installing -- do not interrupt...
/etc/puppet/modules
+-- puppetlabs-stdlib (v4.5.1)
```

2. Docker needs the EPEL repo to be enabled in order to be installed successfully. A quick way to install the EPEL repo is to install the module directly:

```
#puppet module install --force stahnma-epel
Notice: Preparing to install into /etc/puppet/modules...
Notice: Downloading from https://forgeapi.puppetlabs.com...
Notice: Installing -- do not interrupt...
/etc/puppet/modules
+-- stahnma-epel (v1.0.2)
```

3. It's time to install Docker, which is very easy with the excellent Puppet module provided by Gareth R. (*https://github.com/garethr/garethr-docker*). Run the following commands to pull the Docker module from Puppetforge:

```
#puppet module install garethr/docker --ignore-dependencies
Notice: Preparing to install into /etc/puppet/modules...
Notice: Downloading from https://forgeapi.puppetlabs.com...
Notice: Installing -- do not interrupt...
/etc/puppet/modules
+-- garethr-docker (v3.5.0)
[root@puppetmaster ~]#
```

4. Open the */etc/puppet/manifests* folder and create *site.pp* if not available. If newly created, *site.pp* should look like what's shown below:

```
#cat /etc/puppet/manifests/site.pp
node "master.collabnix.com" {
include 'docker'
```

5. Let us apply the Puppet module in order to get Docker installed on the Puppet master:

```
#puppet apply site.pp
Notice: Compiled catalog for master.collabnix.com in
environment production in 1.75 seconds
Notice: /Stage[main]/Docker::Service/File[/etc/
sysconfig/docker-storage]/content: content changed
'{md5}ab8c4963d7da2df915a052babb0e1b89' to '{md5}
a6d7017ae0cf60008ff76ce39f3a4245'
Notice: /Stage[main]/Docker::Service/File[/etc/sysconfig/
docker]/content: content changed '{md5}670c789326a6013f8a4de53
40cb44d95' to '{md5}54701e57c4b50c02936fa990b9c463f5'
Notice: /Stage[main]/Docker::Service/Service[docker]/ensure:
ensure changed 'stopped' to 'running'
Notice: Finished catalog run in 3.04 seconds
```

6. Verify if the Docker installation went fine using the following command:

```
#docker version
Client version: 1.4.1
Client API version: 1.16
Go version (client): go1.3.3
Git commit (client): 5bc2ff8/1.4.1
OS/Arch (client): linux/amd64
```

This installs Docker on the Puppet master through the Docker module.

Setting up the Puppet agents

I assume that Puppet agents are running smoothly and configured to work with the Puppet master. Follow the steps listed below to configure the Puppet agent for installing Docker.

1. Open */etc/puppet/manifests/site.pp* and make an entry for Puppet agents as shown below:

```
node "agent1.collabnix.com" {
include 'docker'
```

2. Run the following command to get Docker configured on the Puppet agents:

```
#puppet agent –t
```

Hence, we have the Puppet master and client configured with Docker installation.

Writing a Puppet manifest to run containers on Docker

Next, it's time to write a Puppet manifest to run containers on Docker. The following Puppet manifest starts the *ajeetraina/collabnix* container:

```
#installation
include 'docker'
#download and install a docker image
docker::image { 'ubuntu':
  image_tag: 'precise'
}
```